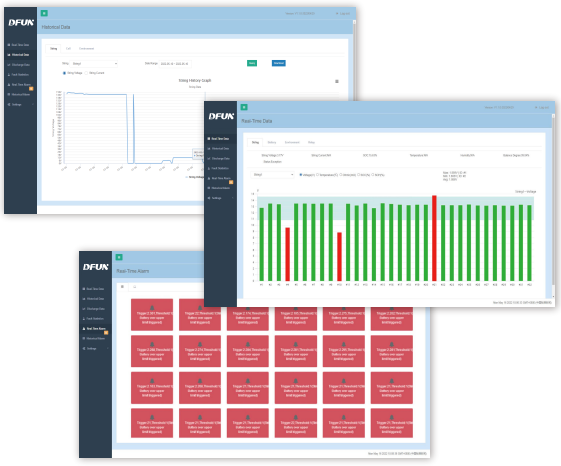
PBMS9000Pro
for Substation



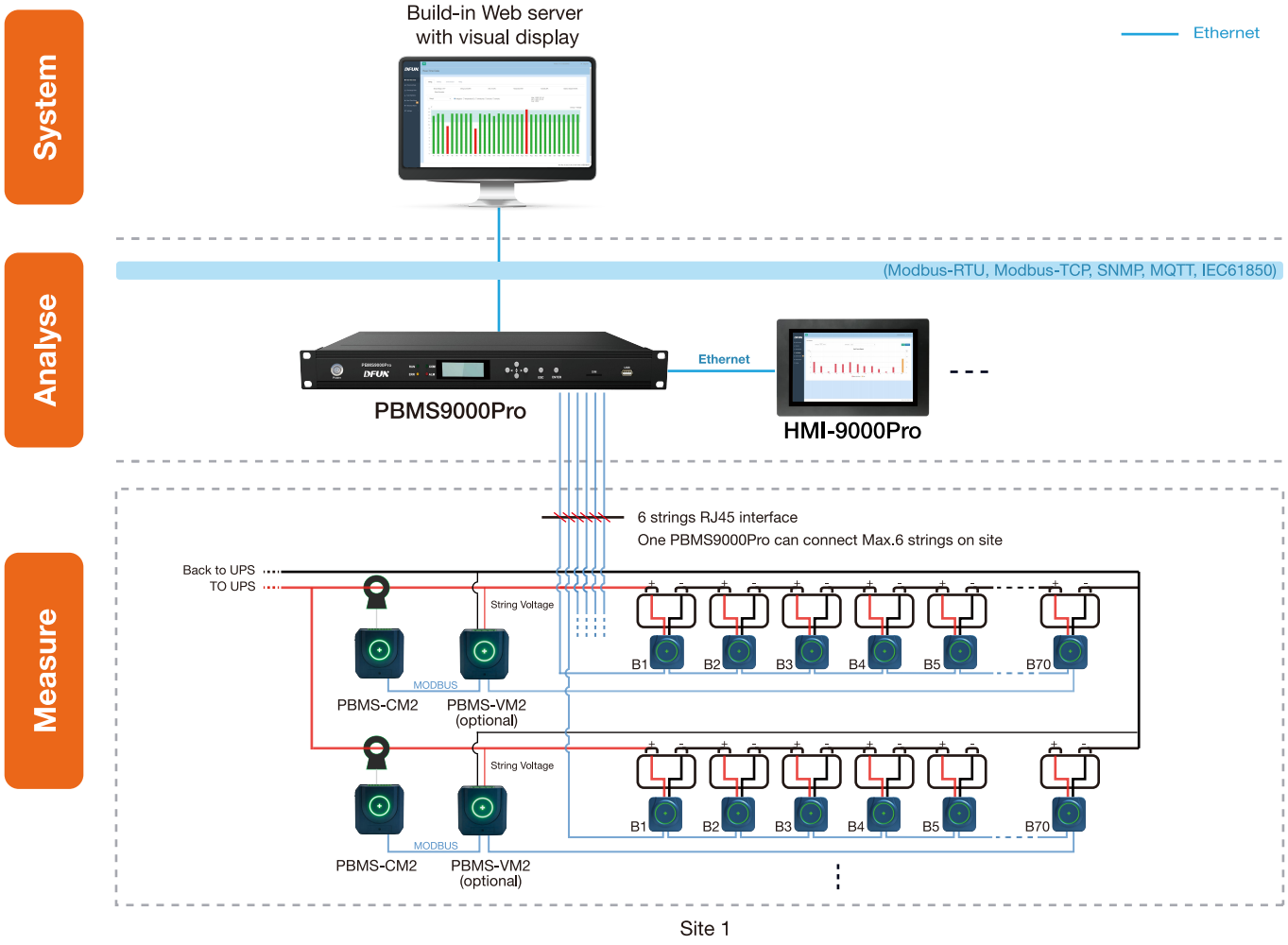
Feature

- Apply to DC power systems like oil & gas, chemical plants, power plants, etc
- 24/7 online monitoring & remote alarm notifications
- Monitor Max. 6 strings and 420pcs batteries in total
- Measure Ni-Cd, Lead-Acid, and multi-pole battery
- Powered by communication bus, no consuming battery power
- Anti-reverse protection, no damaging the cell sensor and battery
- Anti-interference design, support to connect with high-frequency UPS
- Auto-sensing for the battery sensor's ID address
- Support Modbus, SNMP, MQTT and IEC61850 protocols
- Support extension sensor for measuring ambient temperature & humidity, leakage current, insulation resistance (optional)
- Comply with IEEE 1188-2005
- Thermal runaway management(Optional temperature and humidity sensor)



PBMS9000Pro for Substation

System Structure



Site 1

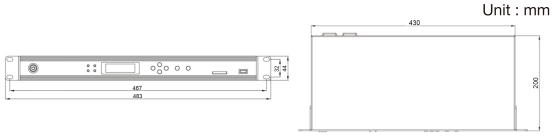
Management Layer

PBMS9000Pro Battery Management Controller

- Standard 1U design for distribution cabinets
- Built-in web server with visual display
- Monitors Max. 6 strings battery, in a total of 420 batteries
- Measures battery string current & voltage, ripple voltage & current, charge & discharge current, internal temperature, impedance, insulation resistance, leakage current, SOC and SOH
- Setpoint alarm for cell voltage, internal temperature, impedance, SOC, SOH (upper/lower limits) and string voltage, current, SOC (upper/lower limits)
- Alarm notifications via SMS or Email
- Supports Modbus-TCP, Modbus-RTU, SNMP and IEC61850 protocols
- Supports MQTT for JSON format to data upload
- 1 RS485 port, 2 Ethernet ports and 1 4G antenna port for data upload
- 6 DI ports (for digital Input connecting)
- 6 DO ports (for sound and light alarms)



Unit : mm

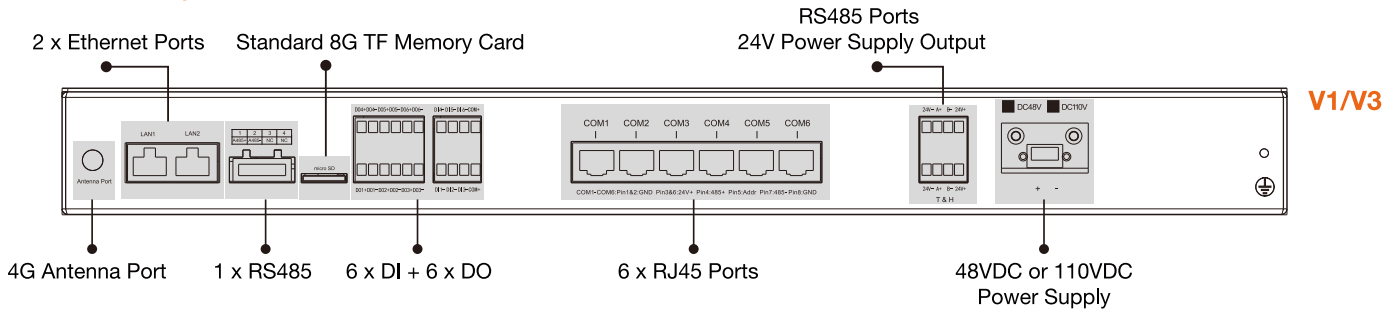


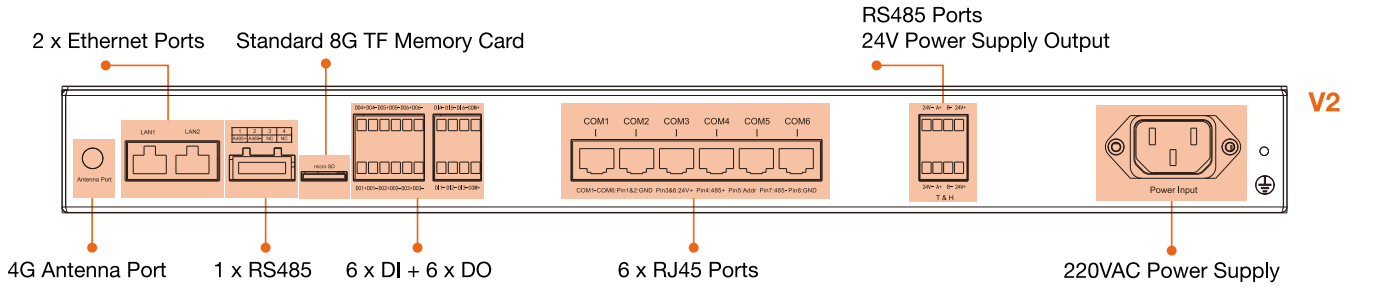
Pre-install & Pre-wiring for PBMS9000Pro and HMI-9000Pro (optional)



PBMS9000Pro for Substation

Interface Description





Technical Specification

CPU	ARM cortex-A7 528MHz	**Up-link Communication**	2 Ethernet ports(10/100M), MODBUS-TCP, SNMP 1RS485 port MODBUS-RTU, baudrate: 9600bps/19200bps/ 38400bps(optional)	
Memory	512MB flash 4G EMMC + 8G TF memory card			
Display	2-inch LCD with backlight	**Down-link Communication**	6 channels RJ45 ports,each port Max. connect 70pcs batteries	
MTBF	≥100,000 hours	**Measure Range**	Voltage	1~6 strings, Range: 20~1000V DC (±0.5%), Resolution: 0.01V
Power Supply	V1: 48VDC, Range: 18~72VDC V2: 220VAC, Range: 90~264VAC/127~370VDC V3: 110VDC, Range: 43~160VDC		Current	1~6 strings, Max. Range: -1000~+1000A DC (±1.0%fs, 15°C~30°C), Resolution: 0.01A
Dimension	Standard 19-inch 1 U device 483mm × 206mm × 44.5mm (W*D*H) Open Hole: 440mm×46mm (L*H)	**Operation Environment**	Operating Temperature: -10~65℃ (±1℃) Storage Temperature: -40~70℃ Humidity: 10~95% RH, non-condensing	
Power Consumption	<15W (Only main controller)			
Additional Port	6 x DI dry contact 6 x DO relay output, 250VAC/5A or 30VDC/5A	**Extension Sensor (Optional)**	-RS485 for Max. 6 strings ambient temperature & humidity measuring -RS485 for 1~6 strings DC leakage current measuring -RS485 for 1~6 strings DC insulation resistance measuring	

String Measuring Sensor

PBMS-CM2 String Current Measuring Sensor & Hall Sensor

- 1 string needs 1 PBMS-CM2, each PBMS-CM2 with 2 hall sensor ports
- Measure battery string charge and discharge current, ripple current
- Hall sensor measuring range: -1000A~+1000A(single-pole) / -2000A~+2000A(dual-pole)
- Supports floating charge hall detection



Hall sensor



Item	Power Supply	Hall Interface	Environment
PBMS-CM2	24VDC (range: 9~32VDC) Power consumption: <0.5W	2 channels, DC ±4V	-10˚C ~ +55˚C , accuracy: ±1˚C

PBMS9000Pro for Substation

Battery Cell Sensor

PBAT71-02 / PBAT71-12 Battery Cell Sensor

- PBAT71-02 for 2V Lead-Acid battery or 1.2V Ni-Cd battery
  PBAT71-12 for 12V Lead-Acid battery
- Monitor individual battery voltage, internal temperature (negative pole), impedance(ohmic value)
- Calculate individual battery state of charge (SOC), state of health (SOH)
- Support battery leakage and liquid level monitoring (optional)
- Powered by communication bus, no consuming battery power
- Auto-balancing



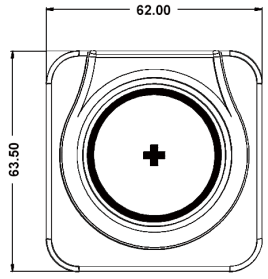
Running Status

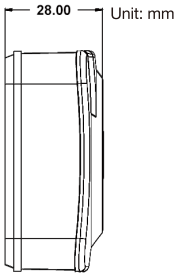


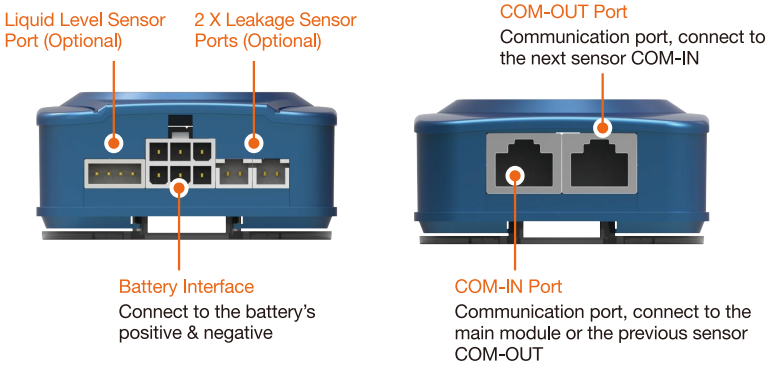
Abnormal Status

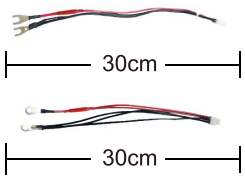
Item	Power Supply	Rated Input Voltage	Measuring Range		
			Voltage	Internal Temperature	Impedance
PBAT71-02	24VDC Power Consumption: < 0.25W	02V / 1.2V	0.5~3VDC (±0.2%)	-20°C~85°C (±0.5°C)	Range: 0.1~50mΩ Repeatability accuracy: ±1.0%±25μΩ Consistency accuracy: ±1.5%±25μΩ
PBAT71-12		12V	5~18VDC (±0.2%)		

Dimension and Installation









	Item	Description
Battery Measuring Cable	Length	30cm
	Terminal & Size	U type, hole diameter: 8mm
		O type, hole diameter: 8mm





PBMS9000Pro for Substation

Extension Module (Optional)

Ambient Temperature & Humidity Sensor



- Dual RJ45 interface, fast wiring, support cascade use
- Adsorption by magnet
- Temperature: -20°C ~ 60°C (±0.4°C), resolution: 0.1°C. Humidity: 0 ~ 100%RH (±3%RH), resolution: 0.1%RH
- Cable(9000-H-THD-RJ45-500): 5 meter RJ45 convert to 4P terminal wire

PBMS-VM2 String Voltage Measuring Sensor



- 1 PBMS-VM2 can measure 1 string's voltage
- Measure battery string voltage and ripple voltage
- Built-in ground fault detection & testing

10.1-inch Android HMI



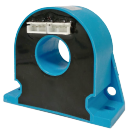
- 10.1" inch touch-screen HMI for local display and operation
- OS: Android 10.0
- CPU: R818 (4-core A53 1.6GHz)
- Running memory: 1GB
- EMMC: 8GB
- Support the reading and configuration of all parameters of the system

Hydrogen (H2) Sensor



- Class of protection: IP65
- Measuring range: 0~1000ppm
- Accuracy: ±5%FS
- Resolution: 1ppm(100KΩ~50KΩ)
- Cable: 2 meter Aviation port to 4P terminal wire

DC Leakage Current Sensor



- Monitoring string leakage current to prevent fire caused
- Cable: 2 meter hall sensor port to 4P terminal wire
- Accuracy: ±1%FS

Order Information

System Structure		Model	Description	Remark
Management Layer	BMS Main Controller	PBMS9000Pro-V1	Battery Management Controller, 48VDC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.100 batteries)
		PBMS9000Pro-V1-L	Battery Management Controller, 48VDC power supply with 4G	
		PBMS9000Pro-V2	Battery Management Controller, 220VAC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.420 batteries)
		PBMS9000Pro-V2-L	Battery Management Controller, 220VAC power supply with 4G	
		PBMS9000Pro-V3	Battery Management Controller, 110VDC power supply	One per UPS One PBMS9000Pro can monitor Max.6 battery strings (Max.216 batteries)
		PBMS9000Pro-V3-L	Battery Management Controller, 110VDC power supply with 4G	
	Protocol(optional)	PBMS9000Pro-CP	DFUN standard IEC61850 model available. For customized needs, please contact our sales team.	
String Measuring Sensor		PBMS-CM2	String current measuring sensor	Each has 2 hall sensor ports, one or two PBMS-CM2 per string Hall sensor cable: 2m cable Communication cable: 5m with RJ45 port
		PBMS-VM2	String voltage measuring sensor	One per each string (optional)
Battery Cell Sensor		PBAT 71-02	02V / 1.2V battery cell sensor	One per battery cell
		PBAT 71-12	12V battery cell sensor	One per battery cell
		PBAT 71-02-B	02V / 1.2V battery cell sensor	One per battery cell,Connector Strip Resistance
		PBAT 71-12-B	12V battery cell sensor	
Accessory	Cell Sensor Communication Cable	PBAT-COM-RJ45-20	Cable terminal: RJ45 port, length: 20cm for Ni-Cd battery	
		PBAT-COM-RJ45-40	Cable terminal: RJ45 port, length: 40cm for Lead-acid battery	
	Battery Measuring Cable	PBAT-C1-U-M8-30	Cable terminal: U type , hole diameter: 8mm, length: 30cm (standard)	
		PBAT-C1-O-M8-30	Cable terminal: O type , hole diameter: 8mm, length: 30cm	

PBMS9000Pro for Substation

System Structure		Model	Description	Remark
Accessory	Hall Sensor for PBMS-CM2	DXEEK03-50A-V4	Rated input: 50A	Measure Range: -50A~+50A, Φ20mm
		DXEEK03-100A-V4	Rated input: 100A	Measure Range: -100A~+100A, Φ20mm
		DXEEK05-25A-V4	Rated input: 25A	Measure Range: -25A~+25A, Φ40mm
		DXEEK05-200A-V4	Rated input: 200A	Measure Range: -200A~+200A, Φ40mm
		DXEEK05-300A-V4	Rated input: 300A	Measure Range: -300A~+300A, Φ40mm
		DXEEK05-400A-V4	Rated input: 400A	Measure Range: -400A~+400A, Φ40mm
		DXEEK05-500A-V4	Rated input: 500A	Measure Range: -500A~+500A, Φ40mm
		DXEEK05-600A-V4	Rated input: 600A	Measure Range: -600A~+600A, Φ40mm
		DXEEK05-800A-V4	Rated input: 800A	Measure Range: -800A~+800A, Φ40mm
		DXEEK05-1000A-V4	Rated input: 1000A	Measure Range: -1000A~+1000A, Φ40mm
Optional Module	HMI for Local Display & Operation	HMI-9000Pro	10-inch touch-screen HMI for local display & operation	Power Supply from Auxiliary power module
	Auxiliary Power Module (choose one out of three)	HDR-30-12	220V AC to 12V DC power supply	Provide power supply for the HMI-9000Pro
		DDR-15L-12	48V DC to 12V DC power supply	
		RSD-30H-12	110V DC to 12V DC power supply	
	DC Leakage Current Sensor	STL8S-MT2-300mA-DF	Rated input: ±300mA	Measure range: -360mA~+360mA, Φ72mm
	Ambient Temperature & Humidity	H-THDRJ45	Temperature: -20~60°C (±0.4°C) Humidity: 0 ~ 100%RH (±3%RH)	1 PBMS9000Pro can conncet Max. 6pcs
	Hydrogen (H_2) Sensor	PBMS-H_2	Measuring range: 0 ~ 1000ppm	One per one PBMS9000Pro
	Liquid Level Sensor	PBMS-ELS	Monitor liquid level, for transparent battery	One per battery, length: 30cm
		PBMS-Y29	Monitor liquid level, for non-transparent battery	One per battery, length: 50cm
	Leakage Sensor	PBMS-LS-O35	Monitor leakage with two ports (connect to the battery +/- port)	One/Two per each battery, length: 30cm Hole diameter: 35mm
		PBMS-LS-O41	Monitor leakage with two ports (connect to the battery +/- port)	One/Two per each battery, length: 30cm Hole diameter: 41mm
	Cabinet for Prewiring In Factory	PBMS-BOX-260	Dimension: 535*260*335mm (W*H*D)	Pre-install & pre-wiring for PBMS9000Pro and HMI-9000Pro
		PBMS-BOX-730	Dimension: 535*730*335mm (W*H*D)	Pre-install & pre-wiring for Max.4 PBMS9000Pro and one HMI-9000Pro
	Ethernet Switch	IES318	1. Power supply 24VDC 12~48VDC 2. 8 x 10/100M Ethernet ports	If selected PBAT-BOX-730, then must select IES318

Note: If battery terminal is M5, M6, M10, M12, M16, please contact our sales for customize requirements before placing the order.

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